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Structural Models

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```
library(lavaan)
library(lavaangui)
library(semPlot)
library(semptools)
library(semTools)
library(ggdag)
library(ggplot2)
library(dplyr)
library(modelsummary)
library(tinytable)
source(here::here("notes", "R", "sem_table.R"))
```

Introduction (Chapter 15)

Structural Regression (SR) models, also known as **Structural Equation Modeling (SEM)** in the broad sense (or "Latent Variable Path Analysis"), combine two components:

1. **Measurement Model**: Confirmatory Factor Analysis (CFA) describing how latent variables are measured by observed indicators.
2. **Structural Model**: Path analysis describing the causal relationships (regressions) among the latent variables.

Causal Assumptions in SR Models

Every directed arrow in the structural model represents a **hypothesized causal effect**, not merely a statistical association. By drawing an arrow from X to Y, the researcher claims that X influences Y—not only that they tend to move together. The structural model is therefore best understood as a **Directed Acyclic Graph (DAG)** extended to include latent variables, and the same principles of causal identification apply. In structural models or causal models, the main challenge is to identify whether variables **not in the model**, either measured or unmeasured, could bias the estimated path coefficients. There are three main types of variables:

Confounder. An unmeasured common cause U of X and Y creates a spurious back-door path:

$$X \leftarrow U \rightarrow Y$$

The estimated path coefficient then conflates the direct effect of X with the spurious association transmitted through U. Blocking this path requires including appropriate observed covariates or using an instrumental variable.

Mediator. Including a variable that lies on the hypothesized causal pathway ($X \rightarrow M \rightarrow Y$) as a covariate when estimating $X \rightarrow Y$ blocks part of X's genuine effect. This is why regression assumptions (no causal order among predictors) cannot be applied uncritically when variables are causally connected.

Collider. Conditioning on a variable caused by two others (a collider, e.g., $X \rightarrow C \leftarrow Y$) opens a spurious association between those two causes. This can arise when selecting participants on a common outcome or when a descendant of a collider is included as a covariate.

The key discipline from causal inference:

> **Identify before you estimate.** Causal assumptions must be justified with subject-matter knowledge before fitting the model. The data alone cannot distinguish a correctly specified causal model from a misspecified one.

When causal structure is uncertain, tools such as `dagitty` can enumerate minimal sufficient adjustment sets and derive the testable conditional independencies implied by the DAG—providing an empirical check on at least some aspects of the assumed structure.

The Two-Step Approach

Many SEM scholars (e.g., Kline, 2023) strongly recommend a two-step modeling strategy:

1. **Respecify and test the measurement model**: specificy a CFA model where all latent variables are allowed to correlate (i.e., saturated structural part).

- Ensure factors are well-measured.
- Check model fit.
- If the measurement model is bad, the SR model will effectively necessarily be bad or misleading.

2. **Test the structural model**: Impose the hypothesized structural paths (regressions) among the latent variables.

Example: Self-Oriented Perfectionism, Self-Efficacy, and Procrastination

This example uses data from Williams & Edwards (2021) to demonstrate a **mediation structural regression model**. The hypothesized model follows Seo (2008):

```
dag <- dagify(
    NGSE ~ SOP,
    PASS ~ NGSE + SOP,
    coords = list(
        x = c(SOP = 0, NGSE = 1, PASS = 2),
        y = c(SOP = 0, NGSE = 1, PASS = 0)
    )
)
ggdag(dag) +
    annotate("text", x = 0.38, y = 0.60,  label = "a",  fontface = "italic",
size = 5) +
    annotate("text", x = 1.62, y = 0.60,  label = "b",  fontface = "italic",
size = 5) +
    annotate("text", x = 1.00, y = -0.12, label = "c'", fontface = "italic",
size = 5) +
    theme_dag()
```

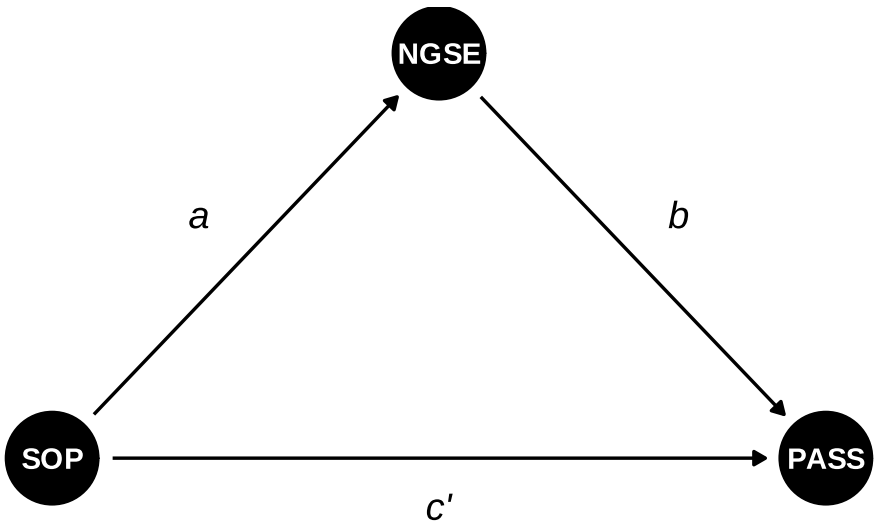

Figure 1: Hypothesized mediation model with path labels.

$$\text{NGSE} = a \cdot \text{SOP} + \zeta_1$$
$$\text{PASS} = b \cdot \text{NGSE} + c' \cdot \text{SOP} + \zeta_2$$

where SOP is self-oriented perfectionism, NGSE is general self-efficacy (Chen et al., 2001), and PASS is academic procrastination.

Data and PASS Parcels

The data are downloaded from OSF using the same exclusion pipeline as the CFA note.

Rather than using all 12 PASS items as indicators of a single factor, Williams & Edwards formed **item parcels**—one sum score per content area:

> ℹ Item Parceling
>
> **Parceling** aggregates multiple items into composite scores before fitting the SEM. Benefits include fewer parameters, a better indicator-to-sample-size ratio, and composites that are more normally distributed than raw ordinal items. The four parcels (assignments, exams, reading, general) replace 12 item indicators with 4 content-balanced composites for the PASS factor.

Descriptive Statistics

```
modelsummary::datasummary_skim(we_data)
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
SOP_1	5	0	3.5	1.1	1.0	4.0	5.0	▂▃▅▃
SOP_2	5	0	3.3	1.2	1.0	3.0	5.0	▃▅▅▅
SOP_3	5	0	3.0	1.1	1.0	3.0	5.0	▃▅▅▂
SOP_4	5	0	3.0	1.2	1.0	3.0	5.0	▃▅▅▂
SOP_5	5	0	2.6	1.2	1.0	2.0	5.0	▅▃▂▂
NGSE_1	5	0	3.9	0.8	1.0	4.0	5.0	▁▁▅▂
NGSE_2	5	0	3.7	1.0	1.0	4.0	5.0	▂▂▅▂
NGSE_3	5	0	4.1	0.8	1.0	4.0	5.0	▁▂▅▃
NGSE_4	5	0	3.9	1.0	1.0	4.0	5.0	▁▂▅▃
NGSE_5	5	0	3.9	0.9	1.0	4.0	5.0	▁▂▅▃
NGSE_6	5	0	3.9	1.0	1.0	4.0	5.0	▁▂▅▃
NGSE_7	5	0	3.7	1.0	1.0	4.0	5.0	▂▅▅▃

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	His-togram
NGSE_8	5	0	3.7	1.0	1.0	4.0	5.0	
Pasgn	13	0	10.9	2.5	3.0	11.0	15.0	
Pexam	13	0	10.8	2.8	3.0	11.0	15.0	
Pread	14	0	10.0	3.3	3.0	10.0	15.0	
Pgen	13	0	10.3	2.6	3.0	10.0	15.0	

Inter-item Correlations

```
nv_we  <- ncol(we_data)
cor_we <- cor(we_data, use = "pairwise.complete.obs")
bg_we  <- rgb(as.integer(cor_we > 0), 0, as.integer(cor_we < 0),
              alpha = abs(cor_we))
bg_we[upper.tri(cor_we, diag = TRUE)] <- NA
modelsummary::datasummary_correlation(we_data) |>
  `colnames<-`(c(" ", seq_len(ncol(we_data)))) |>
  tinytable::format_tt(j = 1, fn = \(x) paste0(seq_len(nv_we), ". ", x)) |>
  tinytable::style_tt(i = 1:nv_we, j = 1:nv_we + 1, background = bg_we) |>
  tinytable::style_tt(fontsize = 0.7)
```

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. SOP_1	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2. SOP_2	.77	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3. SOP_3	.67	.68	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4. SOP_4	.68	.75	.74	1	.	.	.	.	.	.	.	.	.	.	.	.	.
5. SOP_5	.46	.55	.57	.61	1	.	.	.	.	.	.	.	.	.	.	.	.
6. NGSE_1	.24	.14	.18	.15	.05	1	.	.	.	.	.	.	.	.	.	.	.
7. NGSE_2	.18	.10	.20	.13	.07	.53	1	.	.	.	.	.	.	.	.	.	.
8. NGSE_3	.13	.02	.10	.03	-.03	.53	.58	1	.	.	.	.	.	.	.	.	.
9. NGSE_4	.17	.11	.21	.13	.09	.45	.53	.54	1	.	.	.	.	.	.	.	.
10. NGSE_5	.15	.11	.21	.13	.07	.48	.56	.56	.60	1	.	.	.	.	.	.	.
11. NGSE_6	.13	.06	.13	.07	.06	.35	.48	.47	.47	.59	1	.	.	.	.	.	.
12. NGSE_7	.28	.19	.27	.21	.09	.36	.45	.41	.44	.46	.55	1	.	.	.	.	.
13. NGSE_8	.13	.07	.16	.08	.04	.40	.51	.48	.47	.55	.53	.60	1	.	.	.	.
14. Pasgn	.09	-.01	-.10	-.05	.08	-.24	-.22	-.16	-.17	-.15	-.13	-.16	-.17	1	.	.	.
15. Pexam	.08	.02	-.07	-.05	.06	-.17	-.15	-.07	-.05	-.07	-.05	-.08	-.13	.58	1	.	.
16. Pread	.05	.04	-.04	-.01	.03	-.19	-.11	-.12	-.06	-.11	-.07	-.04	-.08	.50	.44	1	.
17. Pgen	.06	-.01	-.06	-.02	.08	-.22	-.17	-.16	-.12	-.10	-.02	-.06	-.10	.64	.54	.51	1

> **Note**
>
> Blue cells = positive correlations, red = negative; intensity = magnitude. SOP items should correlate positively with each other, NGSE items with each other, and the PASS parcels with each other, while cross-scale correlations reflect the hypothesized structural relationships.

Step 1: Measurement Model

We first fit all three factors simultaneously with a saturated structural part (all factors freely correlated) to check that the measurement model is acceptable before imposing the mediation constraints.

```
model_cfa_we <- "
  SOP  =~ SOP_1 + SOP_2 + SOP_3 + SOP_4 + SOP_5
  NGSE =~ NGSE_1 + NGSE_2 + NGSE_3 + NGSE_4 +
          NGSE_5 + NGSE_6 + NGSE_7 + NGSE_8
  PASS =~ Pasgn + Pexam + Pread + Pgen
"
fit_cfa_we <- cfa(
    model_cfa_we,
    data = we_data,
    std.lv = TRUE,
    estimator = "MLR",
    missing = "FIML"
)
summary(fit_cfa_we, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 63 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        54

  Number of observations                           575
  Number of missing patterns                         2

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                               399.684     344.363
  Degrees of freedom                               116         116
  P-value (Chi-square)                           0.000       0.000
  Scaling correction factor                                  1.161
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:
```

```
  Test statistic                              5001.862    4216.065
  Degrees of freedom                               136         136
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.186

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.942       0.944
  Tucker-Lewis Index (TLI)                       0.932       0.934

  Robust Comparative Fit Index (CFI)                         0.945
  Robust Tucker-Lewis Index (TLI)                            0.936

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             -14014.541  -14014.541
  Scaling correction factor                                  1.123
      for the MLR correction
  Loglikelihood unrestricted model (H1)     -13814.699  -13814.699
  Scaling correction factor                                  1.149
      for the MLR correction

  Akaike (AIC)                               28137.081   28137.081
  Bayesian (BIC)                             28372.217   28372.217
  Sample-size adjusted Bayesian (SABIC)      28200.790   28200.790

Root Mean Square Error of Approximation:

  RMSEA                                          0.065       0.059
  90 Percent confidence interval - lower         0.058       0.052
  90 Percent confidence interval - upper         0.072       0.065
  P-value H_0: RMSEA <= 0.050                    0.000       0.017
  P-value H_0: RMSEA >= 0.080                    0.000       0.000

  Robust RMSEA                                               0.063
  90 Percent confidence interval - lower                     0.055
  90 Percent confidence interval - upper                     0.071
  P-value H_0: Robust RMSEA <= 0.050                         0.003
  P-value H_0: Robust RMSEA >= 0.080                         0.000

Standardized Root Mean Square Residual:

  SRMR                                           0.050       0.050

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
```

```
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  SOP =~
    SOP_1             0.899    0.034   26.133    0.000    0.899    0.817
    SOP_2             1.085    0.033   32.710    0.000    1.085    0.872
    SOP_3             0.924    0.034   26.963    0.000    0.924    0.822
    SOP_4             1.051    0.032   32.517    0.000    1.051    0.870
    SOP_5             0.780    0.042   18.452    0.000    0.780    0.651
  NGSE =~
    NGSE_1            0.533    0.042   12.711    0.000    0.533    0.628
    NGSE_2            0.732    0.035   20.668    0.000    0.732    0.740
    NGSE_3            0.582    0.043   13.638    0.000    0.582    0.723
    NGSE_4            0.700    0.042   16.668    0.000    0.700    0.717
    NGSE_5            0.688    0.038   17.966    0.000    0.688    0.782
    NGSE_6            0.663    0.042   15.683    0.000    0.663    0.693
    NGSE_7            0.641    0.040   15.967    0.000    0.641    0.654
    NGSE_8            0.686    0.038   17.915    0.000    0.686    0.708
  PASS =~
    Pasgn             2.039    0.101   20.216    0.000    2.039    0.816
    Pexam             1.951    0.121   16.076    0.000    1.951    0.697
    Pread             2.046    0.120   17.107    0.000    2.046    0.628
    Pgen              2.081    0.104   19.966    0.000    2.081    0.788

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  SOP ~~
    NGSE              0.215    0.053    4.075    0.000    0.215    0.215
    PASS             -0.049    0.046   -1.054    0.292   -0.049   -0.049
  NGSE ~~
    PASS             -0.241    0.047   -5.078    0.000   -0.241   -0.241

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .SOP_1             3.510    0.046   76.509    0.000    3.510    3.191
   .SOP_2             3.259    0.052   62.803    0.000    3.259    2.619
   .SOP_3             2.951    0.047   62.954    0.000    2.951    2.625
   .SOP_4             3.038    0.050   60.307    0.000    3.038    2.515
   .SOP_5             2.633    0.050   52.692    0.000    2.633    2.197
   .NGSE_1            3.859    0.035  109.112    0.000    3.859    4.550
   .NGSE_2            3.650    0.041   88.483    0.000    3.650    3.690
   .NGSE_3            4.077    0.034  121.427    0.000    4.077    5.064
   .NGSE_4            3.857    0.041   94.722    0.000    3.857    3.950
   .NGSE_5            3.923    0.037  106.989    0.000    3.923    4.462
   .NGSE_6            3.885    0.040   97.368    0.000    3.885    4.061
   .NGSE_7            3.680    0.041   89.972    0.000    3.680    3.752
   .NGSE_8            3.743    0.040   92.615    0.000    3.743    3.862
```

```
   .Pasgn            10.941    0.104  105.025    0.000   10.941    4.380
   .Pexam            10.803    0.117   92.595    0.000   10.803    3.861
   .Pread             9.999    0.136   73.566    0.000    9.999    3.069
   .Pgen             10.311    0.110   93.637    0.000   10.311    3.905

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .SOP_1             0.402    0.032   12.716    0.000    0.402    0.332
   .SOP_2             0.371    0.041    9.098    0.000    0.371    0.240
   .SOP_3             0.410    0.038   10.846    0.000    0.410    0.324
   .SOP_4             0.354    0.039    9.181    0.000    0.354    0.243
   .SOP_5             0.827    0.052   15.914    0.000    0.827    0.576
   .NGSE_1            0.436    0.036   12.051    0.000    0.436    0.606
   .NGSE_2            0.442    0.038   11.631    0.000    0.442    0.452
   .NGSE_3            0.309    0.026   12.120    0.000    0.309    0.478
   .NGSE_4            0.464    0.035   13.150    0.000    0.464    0.486
   .NGSE_5            0.300    0.024   12.310    0.000    0.300    0.388
   .NGSE_6            0.476    0.035   13.526    0.000    0.476    0.520
   .NGSE_7            0.550    0.035   15.842    0.000    0.550    0.572
   .NGSE_8            0.468    0.037   12.797    0.000    0.468    0.499
   .Pasgn             2.081    0.253    8.223    0.000    2.081    0.334
   .Pexam             4.020    0.385   10.446    0.000    4.020    0.514
   .Pread             6.425    0.473   13.574    0.000    6.425    0.605
   .Pgen              2.643    0.366    7.217    0.000    2.643    0.379
    SOP               1.000                               1.000    1.000
    NGSE              1.000                               1.000    1.000
    PASS              1.000                               1.000    1.000
```

Step 2: Structural Regression Model

We constrain the structural part to the hypothesized mediation model. Parameter labels (`a`, `b`, `c`) allow us to define the indirect and total effects using the `:=` operator.

```
model_sr_we <- "
  # Measurement model
  SOP  =~ SOP_1 + SOP_2 + SOP_3 + SOP_4 + SOP_5
  NGSE =~ NGSE_1 + NGSE_2 + NGSE_3 + NGSE_4 +
          NGSE_5 + NGSE_6 + NGSE_7 + NGSE_8
  PASS =~ Pasgn + Pexam + Pread + Pgen

  # Structural model (mediation)
  NGSE ~ a * SOP
  PASS ~ b * NGSE + c * SOP

  # Defined parameters
  indirect := a * b
  total    := c + a * b
"
```

```
fit_sr_we <- sem(
    model_sr_we,
    data = we_data,
    std.lv = TRUE,
    estimator = "MLR",
    missing = "FIML"
)
summary(fit_sr_we, fit.measures = TRUE, standardized = TRUE, rsquare = TRUE)
```

```
lavaan 0.6-21 ended normally after 63 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        54

  Number of observations                           575
  Number of missing patterns                         2

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                               399.684     344.363
  Degrees of freedom                               116         116
  P-value (Chi-square)                           0.000       0.000
  Scaling correction factor                                  1.161
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                              5001.862    4216.065
  Degrees of freedom                               136         136
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.186

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.942       0.944
  Tucker-Lewis Index (TLI)                       0.932       0.934

  Robust Comparative Fit Index (CFI)                         0.945
  Robust Tucker-Lewis Index (TLI)                            0.936

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             -14014.541  -14014.541
  Scaling correction factor                                  1.123
      for the MLR correction
  Loglikelihood unrestricted model (H1)     -13814.699  -13814.699
```

```
  Scaling correction factor                                  1.149
      for the MLR correction

  Akaike (AIC)                                28137.081   28137.081
  Bayesian (BIC)                              28372.217   28372.217
  Sample-size adjusted Bayesian (SABIC)       28200.790   28200.790

Root Mean Square Error of Approximation:

  RMSEA                                           0.065       0.059
  90 Percent confidence interval - lower          0.058       0.052
  90 Percent confidence interval - upper          0.072       0.065
  P-value H_0: RMSEA <= 0.050                     0.000       0.017
  P-value H_0: RMSEA >= 0.080                     0.000       0.000

  Robust RMSEA                                                0.063
  90 Percent confidence interval - lower                      0.055
  90 Percent confidence interval - upper                      0.071
  P-value H_0: Robust RMSEA <= 0.050                          0.003
  P-value H_0: Robust RMSEA >= 0.080                          0.000

Standardized Root Mean Square Residual:

  SRMR                                            0.050       0.050

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  SOP =~
    SOP_1             0.899    0.034   26.133    0.000    0.899    0.817
    SOP_2             1.085    0.033   32.710    0.000    1.085    0.872
    SOP_3             0.924    0.034   26.963    0.000    0.924    0.822
    SOP_4             1.051    0.032   32.517    0.000    1.051    0.870
    SOP_5             0.780    0.042   18.452    0.000    0.780    0.651
  NGSE =~
    NGSE_1            0.520    0.041   12.700    0.000    0.533    0.628
    NGSE_2            0.715    0.034   20.754    0.000    0.732    0.740
    NGSE_3            0.568    0.043   13.321    0.000    0.582    0.723
    NGSE_4            0.684    0.042   16.434    0.000    0.700    0.717
    NGSE_5            0.672    0.038   17.712    0.000    0.688    0.782
    NGSE_6            0.648    0.043   15.187    0.000    0.663    0.693
    NGSE_7            0.626    0.039   15.939    0.000    0.641    0.654
    NGSE_8            0.670    0.039   17.386    0.000    0.686    0.708
```

```
  PASS =~
    Pasgn             1.979    0.095   20.873    0.000    2.039    0.816
    Pexam             1.894    0.119   15.981    0.000    1.951    0.697
    Pread             1.986    0.118   16.846    0.000    2.046    0.628
    Pgen              2.019    0.103   19.643    0.000    2.081    0.788

Regressions:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  NGSE ~
    SOP        (a)    0.220    0.057    3.886    0.000    0.215    0.215
  PASS ~
    NGSE       (b)   -0.243    0.052   -4.646    0.000   -0.242   -0.242
    SOP        (c)    0.003    0.049    0.069    0.945    0.003    0.003

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .SOP_1             3.510    0.046   76.509    0.000    3.510    3.191
   .SOP_2             3.259    0.052   62.803    0.000    3.259    2.619
   .SOP_3             2.951    0.047   62.954    0.000    2.951    2.625
   .SOP_4             3.038    0.050   60.307    0.000    3.038    2.515
   .SOP_5             2.633    0.050   52.692    0.000    2.633    2.197
   .NGSE_1            3.859    0.035  109.112    0.000    3.859    4.550
   .NGSE_2            3.650    0.041   88.483    0.000    3.650    3.690
   .NGSE_3            4.077    0.034  121.427    0.000    4.077    5.064
   .NGSE_4            3.857    0.041   94.722    0.000    3.857    3.950
   .NGSE_5            3.923    0.037  106.989    0.000    3.923    4.462
   .NGSE_6            3.885    0.040   97.368    0.000    3.885    4.061
   .NGSE_7            3.680    0.041   89.972    0.000    3.680    3.752
   .NGSE_8            3.743    0.040   92.615    0.000    3.743    3.862
   .Pasgn            10.941    0.104  105.025    0.000   10.941    4.380
   .Pexam            10.803    0.117   92.595    0.000   10.803    3.861
   .Pread             9.999    0.136   73.566    0.000    9.999    3.069
   .Pgen             10.311    0.110   93.637    0.000   10.311    3.905

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .SOP_1             0.402    0.032   12.716    0.000    0.402    0.332
   .SOP_2             0.371    0.041    9.098    0.000    0.371    0.240
   .SOP_3             0.410    0.038   10.846    0.000    0.410    0.324
   .SOP_4             0.354    0.039    9.181    0.000    0.354    0.243
   .SOP_5             0.827    0.052   15.914    0.000    0.827    0.576
   .NGSE_1            0.436    0.036   12.051    0.000    0.436    0.606
   .NGSE_2            0.442    0.038   11.631    0.000    0.442    0.452
   .NGSE_3            0.309    0.026   12.120    0.000    0.309    0.478
   .NGSE_4            0.464    0.035   13.150    0.000    0.464    0.486
   .NGSE_5            0.300    0.024   12.310    0.000    0.300    0.388
   .NGSE_6            0.476    0.035   13.526    0.000    0.476    0.520
   .NGSE_7            0.550    0.035   15.842    0.000    0.550    0.572
```

```
   .NGSE_8           0.468    0.037   12.797    0.000    0.468    0.499
   .Pasgn            2.081    0.253    8.223    0.000    2.081    0.334
   .Pexam            4.020    0.385   10.446    0.000    4.020    0.514
   .Pread            6.425    0.473   13.574    0.000    6.425    0.605
   .Pgen             2.643    0.366    7.217    0.000    2.643    0.379
    SOP              1.000                               1.000    1.000
   .NGSE             1.000                               0.954    0.954
   .PASS             1.000                               0.942    0.942

R-Square:
                   Estimate
    SOP_1             0.668
    SOP_2             0.760
    SOP_3             0.676
    SOP_4             0.757
    SOP_5             0.424
    NGSE_1            0.394
    NGSE_2            0.548
    NGSE_3            0.522
    NGSE_4            0.514
    NGSE_5            0.612
    NGSE_6            0.480
    NGSE_7            0.428
    NGSE_8            0.501
    Pasgn             0.666
    Pexam             0.486
    Pread             0.395
    Pgen              0.621
    NGSE              0.046
    PASS              0.058

Defined Parameters:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
    indirect         -0.054    0.018   -3.040    0.002   -0.052   -0.052
    total            -0.050    0.048   -1.051    0.293   -0.049   -0.049
```

Indirect Effect

The indirect effect (SOP → NGSE → PASS) with its 95% CI from the Monte Carlo method:

```
semTools::monteCarloCI(fit_sr_we, nRep = 20000)
```

```
            est ci.lower ci.upper
indirect -0.054   -0.092   -0.022
total    -0.050   -0.144    0.042
```

Note

The **Monte Carlo method** (MacKinnon et al., 2004) draws repeated samples from the joint sampling distribution of a and b (using their estimated means and covariance matrix) and computes $a \times b$ for each draw, yielding an empirical CI. It handles the asymmetric distribution of a product better than the delta method while requiring no re-estimation of the model, unlike bootstrapping.

Causal Assumptions for the Mediation Interpretation

Reporting $a \times b$ as a causal indirect effect requires that the structural model satisfies a set of **no-unmeasured-confounding** conditions. Three assumptions must hold simultaneously:

1. **No unmeasured confounding of SOP → NGSE**: No latent variable simultaneously causes both self-oriented perfectionism and general self-efficacy.
2. **No unmeasured confounding of SOP → PASS**: No latent variable causes both self-oriented perfectionism and academic procrastination through a path not captured by the model.
3. **No unmeasured confounding of NGSE → PASS** (Sequential Ignorability): No latent variable U simultaneously causes self-efficacy and academic procrastination. This is the most critical assumption—and the hardest to defend. Personality traits, cognitive habits, or prior academic experiences could plausibly affect both NGSE and PASS independently of SOP.

```
dag_u <- dagify(
    NGSE ~ SOP + U,
    PASS ~ NGSE + SOP + U,
    latent = "U",
    coords = list(
        x = c(SOP = 0, NGSE = 1, PASS = 2, U = 1.5),
        y = c(SOP = 0, NGSE = 1, PASS = 0, U = 1.8)
    )
)
ggdag(dag_u) + theme_dag()
```

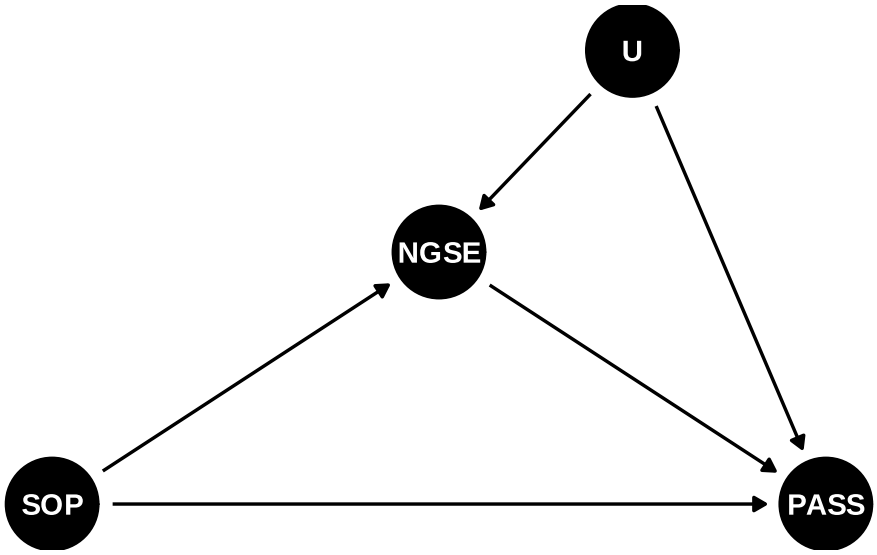

Figure 2: Mediation DAG with a latent confounder U between NGSE and PASS. Sequential ignorability requires that no such U exists.

Because SOP is measured rather than experimentally manipulated, none of these three conditions is guaranteed by design. The indirect effect should therefore be interpreted as a conditional association consistent with the mediation hypothesis, not as a confirmed causal mechanism. A sensitivity analysis—asking how large an unmodeled correlation between the disturbances of NGSE and PASS would need to be to nullify the indirect effect—is a useful addition to any causal mediation claim.

Parameter Estimates Table

```
sem_table(fit_sr_we, caption = "Parameter Estimates for the Structural
Regression Model")
```

Table 1: Parameter Estimates for the Structural Regression Model

	Unstandardized		Standardized		
Parameter	Est	SE	Std	SE_Std	p
Factor loadings					
SOP → SOP_1	.899	.034	.817	.017	< .001
SOP → SOP_2	1.085	.033	.872	.016	< .001
SOP → SOP_3	.924	.034	.822	.019	< .001
SOP → SOP_4	1.051	.032	.870	.016	< .001
SOP → SOP_5	.780	.042	.651	.027	< .001
NGSE → NGSE_1	.520	.041	.628	.035	< .001
NGSE → NGSE_2	.715	.034	.740	.025	< .001

	Unstandardized		Standardized		
Parameter	Est	SE	Std	SE_Std	p
NGSE → NGSE_3	.568	.043	.723	.029	< .001
NGSE → NGSE_4	.684	.042	.717	.027	< .001
NGSE → NGSE_5	.672	.038	.782	.022	< .001
NGSE → NGSE_6	.648	.043	.693	.029	< .001
NGSE → NGSE_7	.626	.039	.654	.029	< .001
NGSE → NGSE_8	.670	.039	.708	.027	< .001
PASS → Pasgn	1.979	.095	.816	.025	< .001
PASS → Pexam	1.894	.119	.697	.033	< .001
PASS → Pread	1.986	.118	.628	.031	< .001
PASS → Pgen	2.019	.103	.788	.031	< .001
Structural coefficients					
SOP → NGSE	.220	.057	.215	.053	< .001
NGSE → PASS	-.243	.052	-.242	.049	< .001
SOP → PASS	.003	.049	.003	.048	0.945

ℹ lavaan syntax

In lavaan:

- `=~` is for measurement (Is measured by)
- `~` is for regression (Is regressed on)
- `~~` is for covariance/variance (Is correlated with)

Visualization

```
lavaangui::plot_lavaan(fit_sr_we)
```

```
p <- semPaths(
    fit_sr_we,
    whatLabels = "std",
    style = "ram",
    layout = "tree2",
    rotation = 2,
    nCharNodes = 0,
    sizeMan = 5,
    sizeLat = 9,
    edge.label.cex = 0.7,
    curvePivot = TRUE,
```

```
    fade = FALSE,
    DoNotPlot = TRUE
)

factor_layout <- matrix(
    c(NA, "NGSE", NA,
      "SOP", NA, "PASS"),
    nrow = 2, byrow = TRUE
)

indicator_push <- c(
    SOP = 1.5,
    NGSE = 1.5,
    PASS = 1.5
)
loading_position <- c(NGSE = 0.7)

p2 <- set_sem_layout(
    p,
    indicator_push  = indicator_push,
    loading_position = loading_position,
    factor_layout    = factor_layout,
    factor_point_to  = c(SOP = "left", NGSE = "up", PASS = "right"),
    indicator_spread = c(SOP = 1.4, NGSE = 2.5, PASS = 1.2)
)

p2 <- mark_sig(p2, fit_sr_we)
```

```
The semPaths plot seems to have one or more intercepts. Support for models with
are only experimental. If failed, consider setting 'intercepts = FALSE' in
semPaths.
```

```
plot(p2)
```

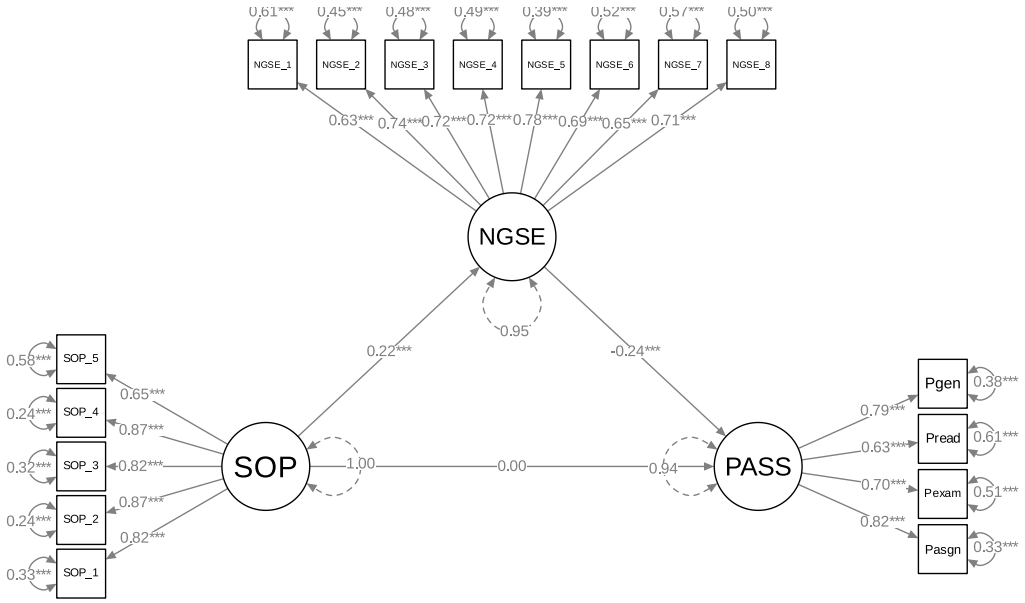

Figure 3: Path diagram from semPaths with semptools styling. Standardized estimates are shown; significant paths are bolded.

Reporting

Sample Write-up

. . .

References

Chen, G., Gully, S. M., & Eden, D. (2001). Validation of a new general self-efficacy scale. Organizational Research Methods, 4(1), 62–83. https://doi.org/10.1177/109442810141004

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MacKinnon, D. P., Lockwood, C. M., & Williams, J. (2004). Confidence limits for the indirect effect: Distribution of the product and resampling methods. Multivariate Behavioral Research, 39(1), 99–128. https://doi.org/10.1207/s15327906mbr3901_4

Seo, E. H. (2008). Self-efficacy as a mediator in the relationship between self-oriented perfectionism and academic procrastination. Social Behavior and Personality, 36(6), 753–764. https://doi.org/10.2224/sbp.2008.36.6.753

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